

SUSSEX, NJ, USA

WMO: 740001

Lat: 41.199N

Lon: 74.626W

Elev: 410

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54793

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.4	(c) 9.0	(d) -8.3	(e) 3.6	(f) 7.4	(g) -3.4	(h) 4.7	(i) 12.6	(j) 17.8	(k) 30.2	(l) 16.2	(m) 30.5	(n) 2.7	(o) 220	(p) 0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.2	(c) 89.5	(d) 73.8	(e) 87.1	(f) 72.4	(g) 84.5	(h) 71.0	(i) 76.6	(j) 85.2	(k) 75.1	(l) 83.1	(m) 73.5	(n) 81.0	(o) 5.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	128.5	80.9	72.5	122.2	79.0	71.3	117.4	77.6	40.2	85.1	38.7	83.3	37.4	81.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
15.7	12.8	11.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-4.5	94.7	8.2	2.5	-10.4	96.5	-15.2	98.0	-19.8	99.5	-25.7	101.3
			-5.1	79.9	8.0	1.5	-10.8	80.9	-15.5	81.8	-20.0	82.6	-25.8	83.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	27.6	29.2	37.6	49.9	60.0	67.9	73.0	70.6	64.0	52.9	41.4	32.9		
	DBStd	17.74	10.08	9.60	9.35	8.37	7.75	5.97	4.68	4.95	6.70	8.21	8.30	8.76		
	HDD50	2665	695	584	399	101	8	0	0	0	0	63	280	535		
	HDD65	5908	1159	1003	849	458	198	39	3	10	99	383	710	997		
	CDD50	2919	1	1	14	98	317	536	713	640	421	154	21	3		
	CDD65	685	0	0	0	5	41	125	251	185	69	9	0	0		
	CDH74	6351	0	0	3	119	523	1188	2294	1539	617	68	0	0		
	CDH80	1942	0	0	0	31	150	342	802	445	162	10	0	0		
	WSAvg	4.0	4.6	4.9	5.5	5.3	3.9	3.2	2.8	2.5	2.9	3.9	4.1	4.4		
	PrecAvg	45.0	3.4	2.6	3.8	3.5	3.5	4.5	4.0	4.3	4.2	3.9	3.3	3.9		
(15) Precipitation	PrecMax	67.2	7.8	7.1	8.7	7.6	7.8	9.8	7.3	13.0	11.3	14.5	5.1	7.9		
	PrecMin	27.0	1.6	0.7	0.8	1.7	0.9	0.9	1.1	1.3	0.1	0.0	1.3	1.1		
	PrecStd	9.5	1.5	1.3	1.9	1.5	1.6	2.3	1.5	2.4	3.1	3.2	1.1	1.7		
	PrecStd	9.5	1.5	1.3	1.9	1.5	1.6	2.3	1.5	2.4	3.1	3.2	1.1	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	62.9	72.5	84.5	88.3	91.2	93.8	91.0	89.0	80.6	69.7	61.5		
		MCWB	56.9	54.2	57.4	62.0	70.4	74.7	74.8	76.1	73.5	67.2	59.2	57.8		
	2%	DB	52.1	54.1	65.1	77.0	84.4	87.3	90.0	87.7	84.8	75.0	64.0	55.4		
		MCWB	47.7	47.0	52.7	59.3	68.1	72.2	74.4	73.7	70.8	65.0	56.6	51.5		
	5%	DB	45.4	47.7	57.7	71.8	80.2	83.8	87.3	84.6	80.8	71.4	59.9	50.0		
		MCWB	41.3	41.1	47.9	57.3	65.6	70.3	73.0	71.8	68.6	63.6	53.0	46.1		
	10%	DB	41.2	42.7	52.7	66.5	75.4	81.0	84.6	82.1	77.3	67.4	55.5	46.0		
		MCWB	37.1	37.3	45.3	53.4	63.0	68.6	71.7	70.5	67.3	60.1	50.0	41.8		
	0.4%	WB	57.6	57.3	59.7	66.6	73.4	76.5	78.4	79.2	76.6	71.7	63.2	58.9		
		MCDB	59.8	62.3	67.5	76.1	84.5	86.8	88.6	88.2	84.8	76.4	67.5	61.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.5	48.3	54.7	62.9	70.8	74.6	76.7	76.4	73.7	68.5	58.6	52.2		
		MCDB	51.3	53.4	63.2	74.8	80.9	83.7	85.9	83.8	80.2	72.9	62.8	54.7		
	5%	WB	41.8	41.8	50.0	58.9	68.3	72.8	75.2	74.3	71.4	65.3	54.8	46.5		
		MCDB	44.9	46.7	56.2	68.3	76.4	81.2	83.4	81.1	77.8	70.1	57.8	49.3		
	10%	WB	37.8	37.6	45.9	55.2	65.8	70.7	73.8	72.5	69.4	61.4	50.6	42.0		
		MCDB	40.5	41.8	52.4	63.1	72.8	77.6	81.4	78.8	74.3	66.1	55.1	45.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	18.5	19.2	22.6	22.7	21.4	21.2	21.0	21.9	20.7	19.0	15.3		
		MCDBR	21.6	25.4	27.5	31.9	28.8	25.4	24.6	23.4	25.8	25.3	25.0	19.7		
		MCWBR	18.0	18.9	17.0	16.4	14.5	12.3	11.1	11.1	13.3	15.2	17.9	17.2		
		MCDBR	20.8	23.4	23.9	26.2	23.7	22.2	20.4	20.0	20.1	20.4	20.7	18.1		
	5% WB	MCWBR	18.3	18.5	16.9	16.1	13.3	11.8	10.1	10.3	11.5	14.2	17.6	17.3		
		MCWBR	18.3	18.5	16.9	16.1	13.3	11.8	10.1	10.3	11.5	14.2	17.6	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.320	0.350	0.370	0.414	0.475	0.479	0.463	0.382	0.356	0.321	0.301		
	taud		2.430	2.376	2.381	2.370	2.275	2.137	2.141	2.176	2.375	2.458	2.514	2.505		
	Ebn at Noon		271	282	285	286	274	257	254	255	271	266	263	263		
	Edh at Noon		25	31	34	37	42	48	48	45	34	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		563	843	1144	1464	1634	1776	1841	1595	1331	905	629	465		
	RadStd		44	88	92	151	170	165	100	104	151	100	61	49		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+115	

Nomenclature: See separate page